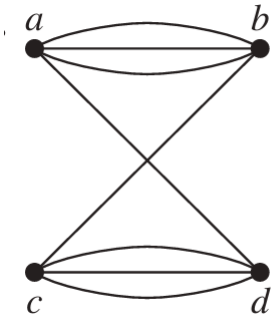
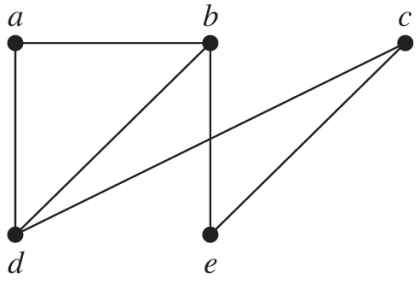
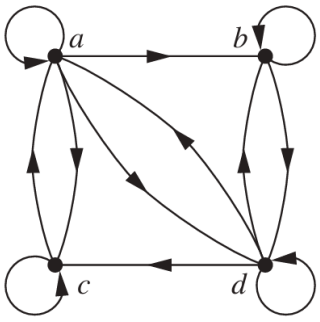
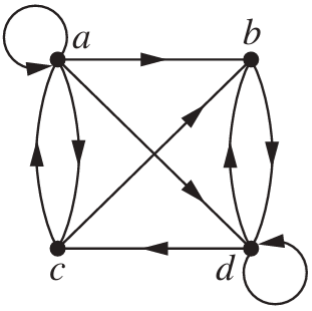
Tutorial 5: Graph Theory II

Question 1: Represent the graph with an adjacency matrix.









Question 2: Draw a graph with the given adjacency matrix

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

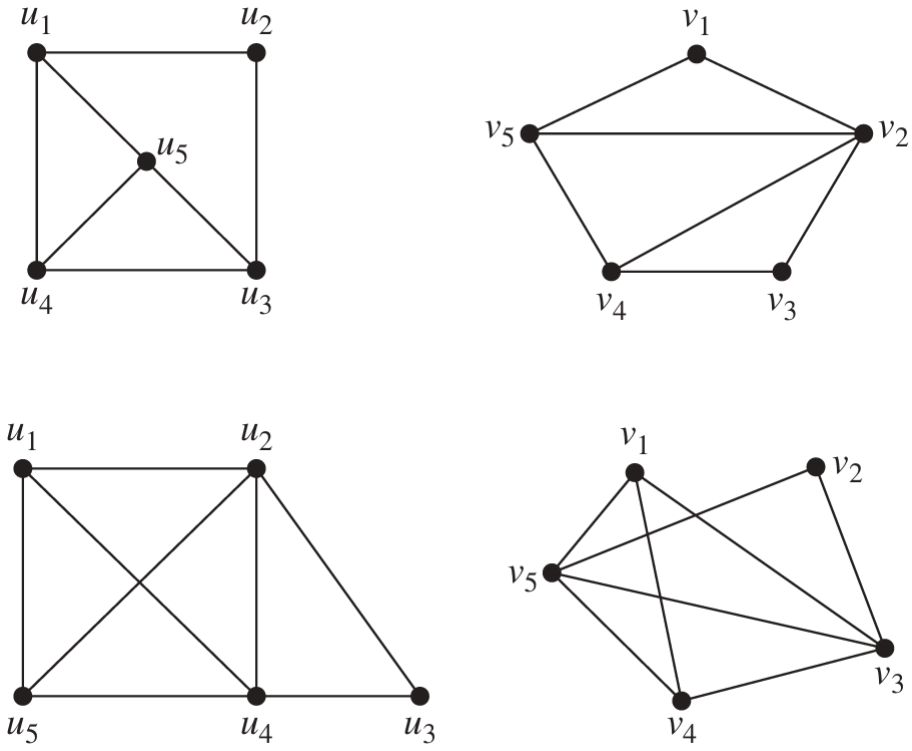
$$\begin{bmatrix} 0 & 2 & 3 & 0 \\ 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Question 3: Is every zero–one square matrix that is symmetric and has zeros on the diagonal the adjacency matrix of a simple graph?

Question 4: What is the sum of the entries in a row of the adjacency matrix for an undirected graph? For a directed graph? What is the sum of the entries in a column of the adjacency matrix for an undirected graph? For a directed graph?

Question 5: Find an adjacency matrix for K_n and C_n

Question 6: Determine whether the given pair of graphs is isomorphic.

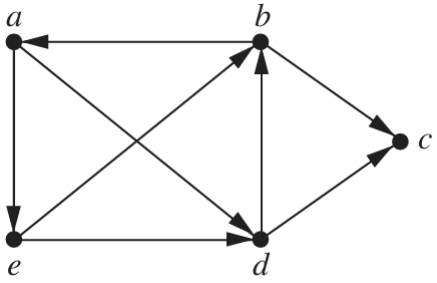


Question 7: Are the simple graphs with the following adjacency matrices isomorphic?

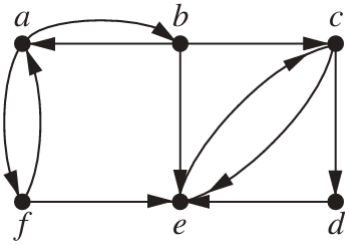
a) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

Question 8: Determine whether each of these graphs is strongly connected and if not, whether it is weakly connected.



Question 9: Find the strongly connected components of each of these graphs.



Question 10: Show that every connected graph with n vertices has at least n-1 edges.